Land Risk Intelligence

An AI-Powered Geospatial Risk Assessment Platform

Team Members

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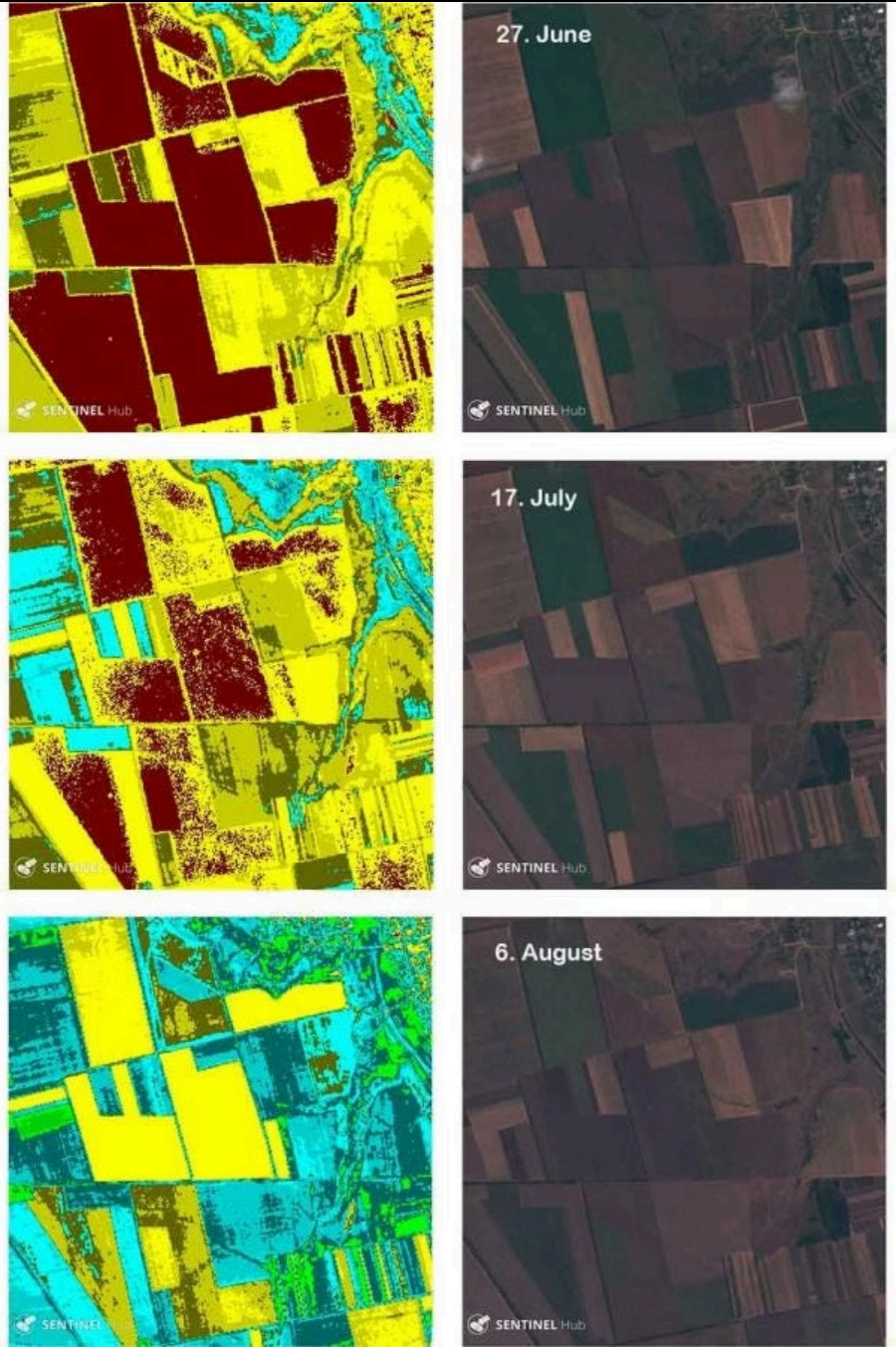
Submission

Machine Learning Project



Problem Statement & Objective

The Challenge

Slow & Complex Assessment

Land eligibility and compliance checks require manual review across multiple regulatory layers, causing delays.

Fragmented Data Sources

Multiple heterogeneous geospatial datasets must be reconciled, often containing inconsistent or incomplete records.

Lack of Explainability

Existing tools produce raw predictions without traceable reasoning, undermining stakeholder trust and compliance.

Our Objective

Build an **AI-powered platform** that automates end-to-end land risk assessment — integrating geospatial data ingestion, machine learning inference, and rule-based validation — to generate **fast, explainable, and auditable risk reports**.

Target: Reduce assessment time from days to seconds with traceable, defensible outputs.

System Architecture

A modular, microservices-based architecture ensuring scalability, fault isolation, and independent deployability of each service layer.

Frontend

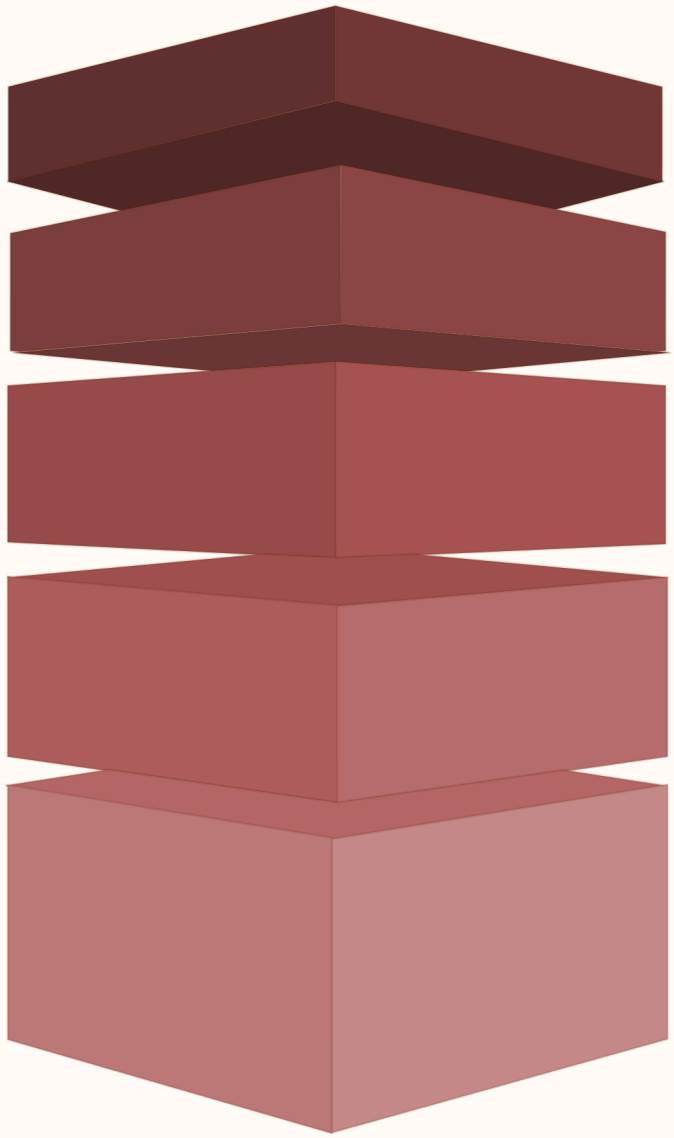
React + Leaflet interactive map UI.

Data & ML

KGIS ingestion + Python XGBoost inference.

Storage

PostgreSQL for persistence; Redis caching.



Backend API

Go REST endpoints with routing.

Validation

Rule-based safety and override engine.

Microservices
Each service is independently deployable and maintainable.

Scalable Design
Horizontal scaling supported via Docker Compose orchestration.

Working Pipeline & Data Flow

Pipeline Highlights

Real-Time Feedback

Live activity progress panel updates users at each pipeline stage, eliminating perceived latency.

Low-Latency Execution

Redis caching accelerates repeated queries by serving pre-computed results instantly.

Traceable Audit Logs

Every pipeline step is logged, enabling full reproducibility and compliance audit trails.

Technologies Used

Frontend

React + **Vite** for a fast, component-driven UI. **Leaflet.js** for interactive geospatial map rendering.

Backend

Go (Golang) for high-performance, concurrent REST API handling with minimal memory overhead.

ML Service

Python with **XGBoost** for gradient-boosted geospatial risk classification and feature importance.

Data Layer

PostgreSQL for persistent geospatial storage. **Redis** for sub-millisecond query caching.

Deployment

Docker Compose for container orchestration. **Nginx** as reverse proxy and load balancer.

Key Features & Innovation

Land Risk Intelligence introduces a differentiated feature set that advances beyond conventional GIS tools toward intelligent, user-centered risk analysis.

AI Risk Scoring

XGBoost-driven geospatial risk quantification with feature importance outputs.

Explainable Reports

Human-readable reports with traceable reasoning for every risk decision rendered.

Evidence Fusion

Satellite imagery and vector map data are fused for richer, multi-source feature sets.

Safety Overrides

Rule-based validation layer enforces hard compliance boundaries independent of the ML model.

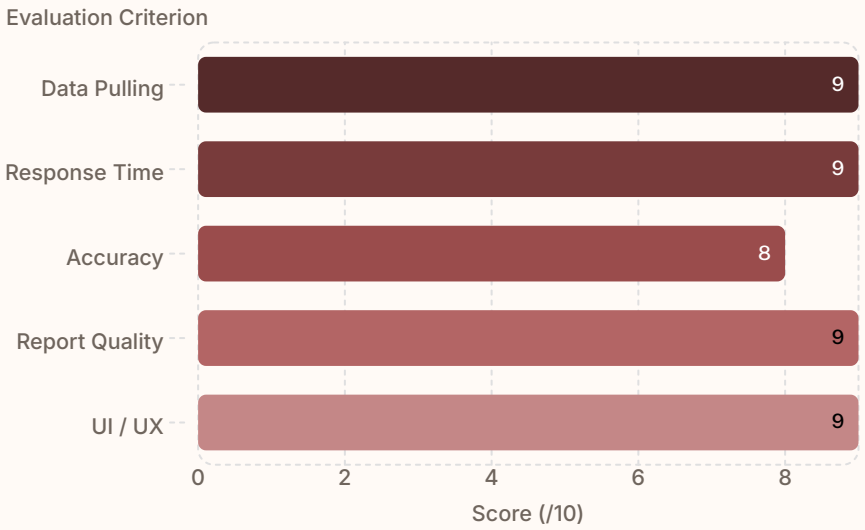
Redis Caching

Repeated geospatial queries served from cache, drastically reducing end-to-end latency.

Audit Logs

Full trace logs for debugging, reproducibility, and regulatory auditability of every assessment.

Results & Evaluation



Performance Outcomes

Stable End-to-End Performance

Full pipeline executed reliably across varied geospatial inputs without timeouts or data loss.

Accelerated Repeated Queries

Redis cache reduced latency by serving prior results without re-triggering ML inference.

Increased User Trust

Live progress indicators and explainable outputs measurably improved perceived reliability.

ML + Validation Accuracy

Combined XGBoost inference with rule-based overrides yielded consistent, defensible results.

Conclusion & Future Scope

Conclusion

Land Risk Intelligence successfully unifies AI inference, geospatial analytics, and scalable cloud architecture into a single platform — transforming a slow, manual assessment process into a **fast, explainable, and auditable** automated pipeline.

Demonstrates strong alignment with real-world enterprise and government deployment requirements.

Future Scope

01

Live Stage Streaming

WebSocket-based real-time pipeline progress for sub-step granularity.

02

Model Calibration

Expanded training datasets and probabilistic calibration for improved accuracy.

03

Additional Satellite Sources

Integration with Sentinel-2, MODIS, and open imagery APIs.

04

Mobile & Enterprise Deployment

Native mobile app and government-grade enterprise licensing roadmap.

Thank You — We welcome questions and feedback from the evaluation panel.